



2026

阀产品手册

Valve Product Catalog >>>



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目录 Contents



01

公司简介
Company Profile

03

暖通电子膨胀阀
命名规则
HVACR EEV
NAMING RULES

04

暖通电子膨胀阀
产品系列型谱
PRODUCT SERIES LINEUP
OF HVACR EEV

05

M系列电子膨胀阀
M Series EEV

09

W系列电子膨胀阀
W Series EEV

13

N系列电子膨胀阀
N Series EEV

17

L系列电子膨胀阀
L Series EEV

21

车用阀命名规则
Vehicle Valve
Naming Rules

22

车用阀产品系列型谱
Product Series Lineup of
Vehicle Valve

23

C1 系列电子膨胀阀
C1 Series EEV

26

D1 系列电子膨胀阀
D1 Series EEV

29

E1 系列电子膨胀阀
E1 Series EEV

32

D2 系列电子膨胀阀
D2 Series EEV

35

H1 系列电磁阀
H1 SERIES
SOLENOID VALVES

38

H2 系列电磁阀
H2 SERIES
SOLENOID VALVES

41

H3 系列电磁阀
H3 SERIES
SOLENOID VALVES

43

Q1 系列单向阀
Q1 SERIES
CHECK VALVE

45

S1 系列单向阀
S1 SERIES
CHECK VALVE

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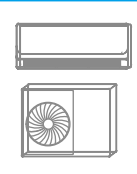
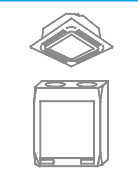
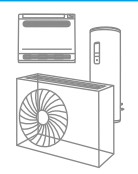
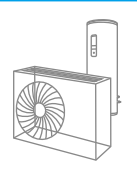
GMCC & Welling & MR are integrated core component solution suppliers for HVACR and appliance, adheres to green innovation, continuously upgrades core technology, provides customers with reliable, green, efficient and intelligent products and solutions such as all kinds of compressors, valves, motors and chips, and helps the healthy development of the industry and the continuous upgrading of the industry.

我们的产品应用 | Product

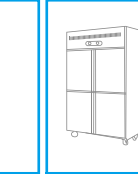
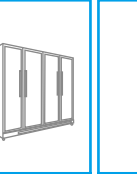
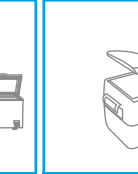
产品广泛应用于暖通家用电器、商用制冷制热和移动制冷等领域，如各类空调、冰箱、洗衣机、冷冻冷藏柜、热泵采暖、热泵热水器、除湿机、干衣机、饮水机、油烟机、洗碗机、吸尘器、储能冷却系统、机房空调、商用车电池热管理、冷藏汽车、驻车空调、房车空调、车载冰箱等。

The products are widely used in HVAC household appliances, commercial refrigeration and heating, mobile refrigeration and other fields, such as various air conditioners, refrigerators, washing machines, freezers and refrigerated cabinets, heat pump heating, heat pump water heaters, dehumidifiers, dryers, water dispensers, range hoods, dishwashers, vacuum cleaners, energy storage cooling systems, computer room air conditioners, commercial vehicle battery thermal management, refrigerated trucks, parking air conditioners, RV air conditioners, vehicle-mounted refrigerators, etc.

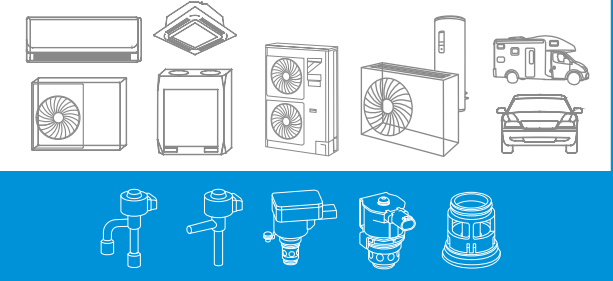
GMCC 转子&涡旋压缩机
Rotary & Scroll Comp.

 家用空调压缩机 Household A/C Comp.	 商用空调压缩机 Commercial A/C Scroll Comp.	 除湿机压缩机 Dehumidifier Comp.	 热泵采暖压缩机 HP Heating Comp.	 热泵热水器压缩机 HP Water Heater Comp.	 热泵干衣机压缩机 HP Dryer Comp.	 微型压缩机 Micro Comp.	 车载空调压缩机 Car A/C Comp.
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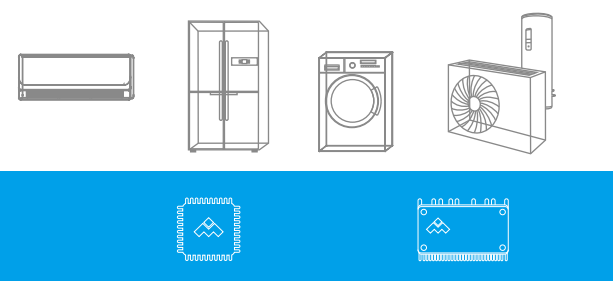
GMCC 往复式压缩机
Recip Comp.

 家用冰箱压缩机 Household Refrigerator Comp.	 家用冷柜压缩机 Household Freezer Comp.	 除湿机压缩机 Dehumidifier Comp.	 轻商设备压缩机 Commercial Comp.	 车载冰箱压缩机 Car Refrigerator Comp.	 房车冰箱压缩机 RV Refrigerator Comp.
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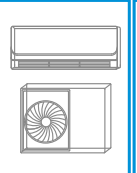
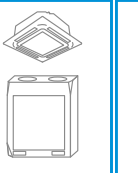
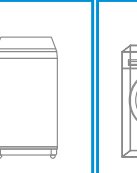
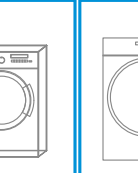
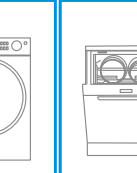
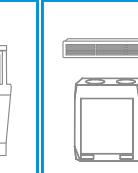
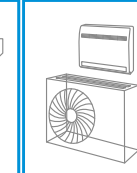
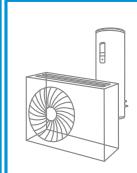
GMCC 阀产品
Valve

 阀 Valve

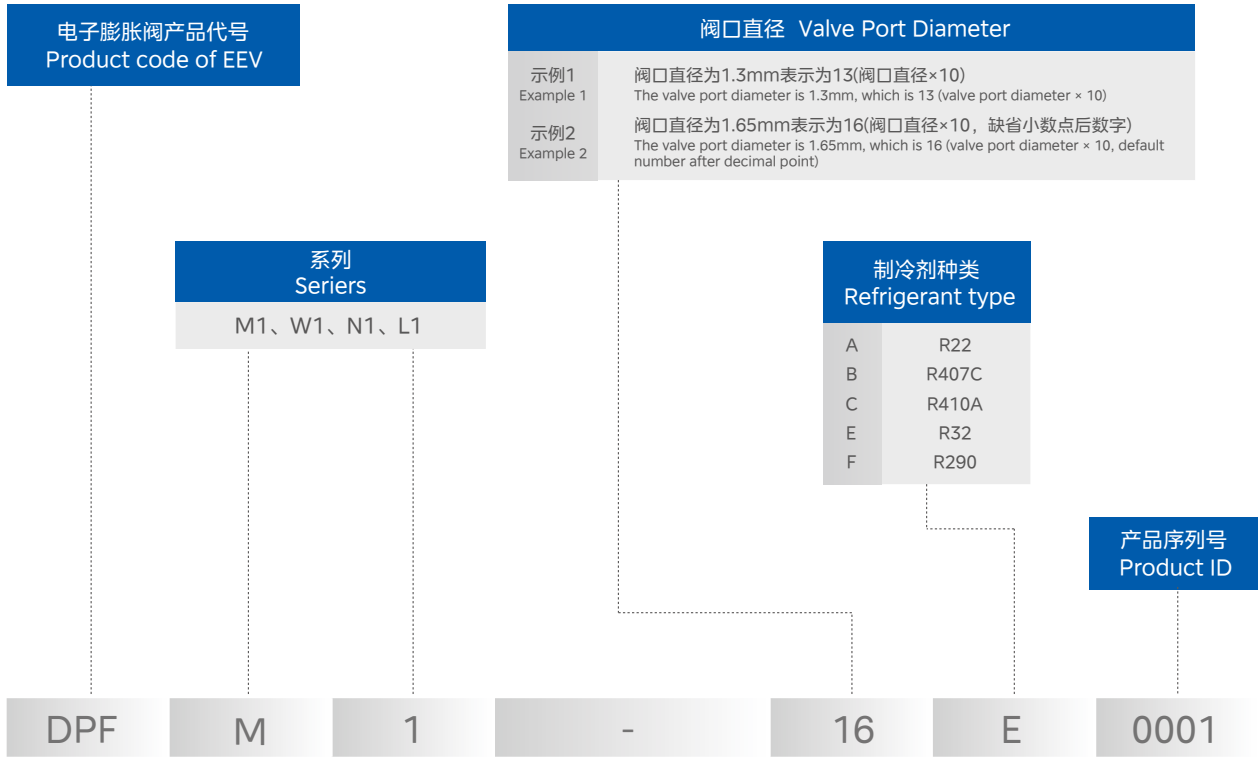
美仁芯片
MR chip

 主控、触控、变频MCU Control, Touch, Motor Control MCU 智能功率模块 IPM
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Welling 电机
Motor

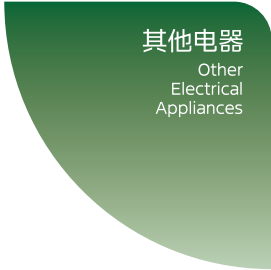
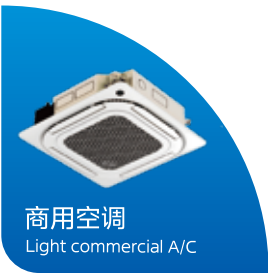
 家用空调电机 Household A/C Motor and Fan	 中央空调电机 Commercial A/C Motor	 波轮洗衣机电机 Top-loading Washing Machine Motor	 滚筒洗衣机电机 Front-loading Washing Machine Motor	 干衣机电机 Dryer Motor	 洗碗机电泵 Dishwasher Electric Pump	 中央空调排水泵 Central A/C Drainage Pump	 热泵采暖循环泵 HP Heating Circulating Water Pump	 热泵热水EC风机 HP Water Heater EC Fan
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暖通电子膨胀阀命名规则
HVACR EEV Naming Scheme



暖通电子膨胀阀产品系列型谱
Product Series Lineup of HVACR EEV

应用领域 Application Area	应用场景 Application Scenario	系列 Series	口径 Port Diameter (mm)	名义容量Theoretical capacity (kW)/R410A													
				2.5kW	3.5kW	5.0kW	7.2kW	12.5kW	20kW	25kW	35kW	50kW	60kW	90kW	120kW	180kW	400kW
家用空调 一拖一 Household A/C One by one	外机 Outdoor unit	M	1.0~3.2														
	内机 Indoor unit	W	1.3~3.2														
商用 Light Commercial A/C 多联机 Multiple on-line 热泵 HP	外机 Outdoor unit	M	1.0~3.2														
	外机 Outdoor unit	N	3.5~6.5														
	外机 Outdoor unit	L	8~13														
商用 Commercial A/C 热泵HP 大型商用 Commercial A/C 可燃冷媒机组 双向截断 Bi-Directional Shut-Off for flammable refrigerant unit																	

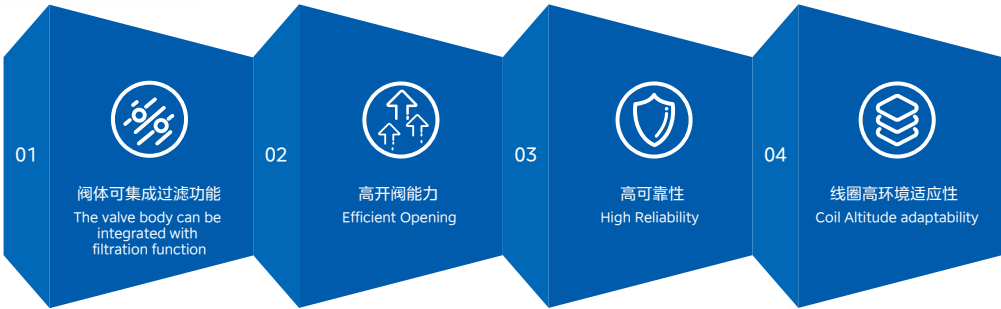




产品应用与特点
Product application and features

M系列电子膨胀阀主要应用于家用空调以及热泵等系统外机中，实现制冷剂流量的自动调节，使系统在最佳工况下运行，快速制冷或制热，精确控温和节能。

M series electronic expansion valve is mainly used in householdair conditioning and heat pump and other systems, to realize the automatic adjustment of refrigerant flow.Make the system run in the best condition, fast cooling or heating, accurate control and energysaving.



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R410A、R32、R290、R22、R407C、R454B、R314a、R507等
◆ 容量范围 Capacity Range	40kW及以下（R410A名义容量） 40kW MAX（R410A Nominal capacity）
◆ 冷媒温度 Refrigerant Temperature	-40℃~+85℃（通电率50%以下） -40℃~+85℃（duty ratio 50% MAX）
◆ 环境温度 Ambient Temperature	-30℃~+60℃（通电率50%以下） -30℃~+60℃（duty ratio 50% MAX）
◆ 环境湿度 Ambient Humidity	≤95%RH
◆ 介质流向 Flow Direction	正反皆可 Bi-flow
◆ 全开脉冲 Full Stroke Pulses	500P
◆ 开阀脉冲 On-valve Pulses	32±20P
◆ 安装方式 Installation Method	以阀体及线圈的断面中心线为轴，并且将线圈朝上，在垂直±15°以内 Take the center line of the valve body and coil section as the axis, and turn the coil up, within the vertical ±15°

电气参数 ELECTRICAL PARAMETER		
◆ 额定电压	Rated Voltage	DC12V±10%，矩形波 DC12V±10%,Rectangular wave
◆ 驱动方式	Actuating Mechanism	直动式4相8拍永磁步进电机 4-phase 8-beat permanent magnet stepper motor,direct-acting
◆ 励磁方式	Recommended Drive Condition	1~2相励磁，单极驱动 1~2 phase excitation,Unipolar drive
◆ 励磁速度	Excitation Rate	Φ1.0 ~ Φ2.4mm :30 ~ 90PPS Φ2.5 ~ Φ3.2mm :30 ~ 40PPS
◆ 额定电阻	Rated Resistance	46±3.7Ω/ 相 (20℃) 46±3.7Ω/ phase(20℃)
◆ 额定电流	Rated Current	0.26A/ 相 (20℃) 0.26A/ phase(20℃)
◆ 绝缘等级	Insulation Class	E级 Class E
◆ 防护等级	Protection Class	IP67

技术参数
Technical Parameter

系列 Production Series	口径 Port Diameter	名义容量(kW) Nominal Capacity						最大工作压力 MAX Operating Pressure	最大动作压差 Max Operating Pressure Difference	逆向开阀压差 Converse On-valve Operating Pressure Difference
-	mm	R410A	R32	R290	R22	R134a	R407C	MPa	MPa	MPa
DPFM*-10*	1.0	4.4	6.5	3.9	3.8	3.0	4.0	4.9	3.5	≥2.1
DPFM*-13*	1.3	6.1	9.2	5.6	5.2	4.1	5.4			
DPFM*-16*	1.65	10.4	15.5	9.4	8.9	6.9	9.2			
DPFM*-18*	1.8	12.4	18.4	11.1	10.6	8.3	10.9			
DPFM*-20*	2.0	15.3	22.8	13.7	13.1	10.2	13.5			
DPFM*-22*	2.2	16.5	24.5	14.8	14.1	11.0	14.5			
DPFM*-24*	2.4	19.4	28.8	17.4	16.6	12.9	17.1			
DPFM*-30*	3.0	32.5	47.7	28.8	27.8	21.7	28.6			≥1.5
DPFM*-32*	3.2	35.5	52.6	31.7	30.3	23.7	31.3			

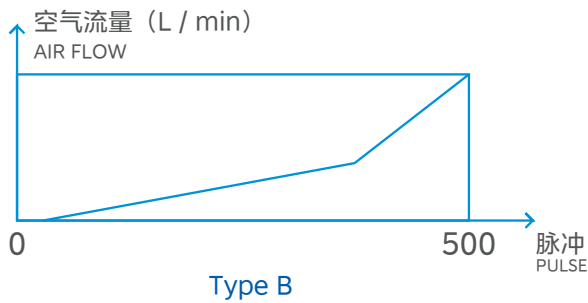
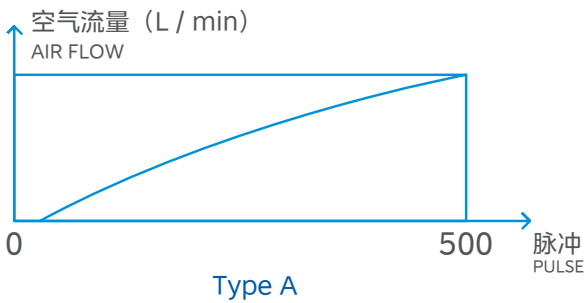
注：额定工况：冷凝温度38℃；蒸发温度5℃；过冷度0℃；过热度0℃。
Rated working condition: condensing temp. 38 °C, evaporating temp. 5 °C, sub-cooled 0 °C, overheat 0 °C.
全关有流量产品无逆向开阀压差
The valve which is completely closed with no flow have no converse on-valve operating pressure difference.

流量特点

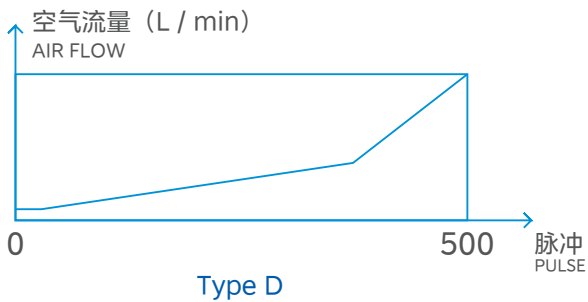
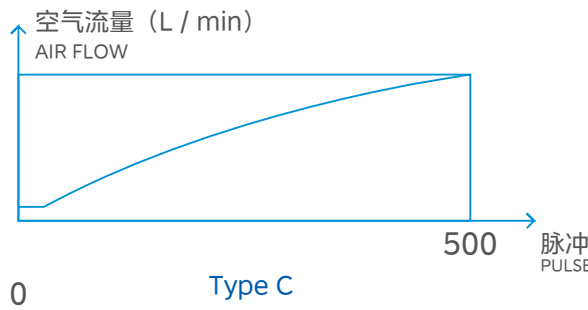
Flow Characteristics

类型 type	A	B	C	D
流量曲线 Flow Curve	线性 Linear	非线性 Inflection	线性 Linear	非线性 Inflection
阀全闭 Valve Full Close	有 Yes	有 Yes	无 None	无 None

M1 SERIES M1系列



M2 SERIES M2系列

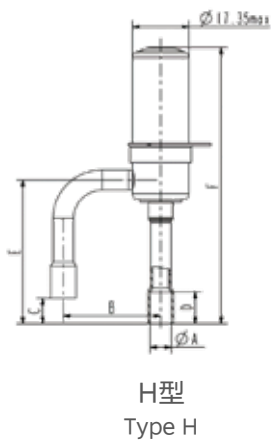


外形尺寸

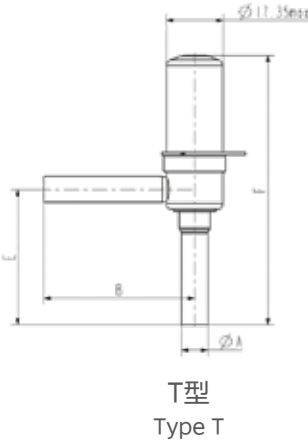
External Dimension

阀体

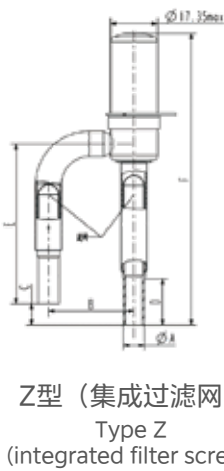
VALVE BODY



H型
Type H



T型
Type T

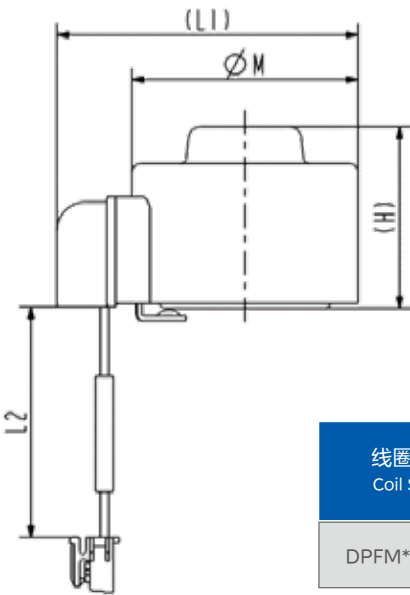


Z型 (集成过滤网)
Type Z
(integrated filter screen)

外形 Shape	系列 Series	基本尺寸 (mm) Basic size of valve					
		A	B	C	D	E	F
H型 Type H	DPFM*-10~24*	6.5	30	8	10	45	86.5
T型 Type T	DPFM*-10~32*	7.94	30	/	/	45	86.5
Z型 Type Z	DPFM*-10~32*	6.5	31.5	8	12	63	104.5

COIL

线圈



线圈系列 Coil Series	线圈基本尺寸 (mm) Basic Size of Coil				连接器型号 Housing Type
	H	M	L1	L2	
DPFM*-10~32*	30.5	38.5	51	750	XHP-5
					XHP-6

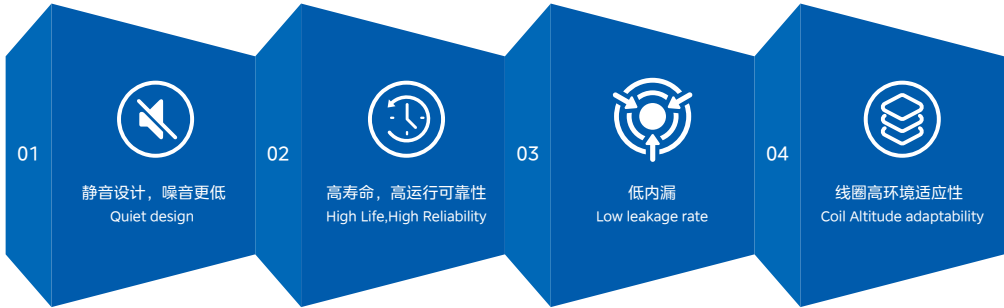
* 线圈导线长度及连接器型号可定制 Wire length and housing type can be customized



产品应用与特点
Product application and features

W系列电子膨胀阀主要应用于轻商空调、多联机和热泵等系统内机及外机中，实现制冷剂流量的自动调节，使系统在最佳工况下运行，快速制冷或制热，精确控温和节能。

W series electronic expansion valve is mainly used in light commercial air conditioning, multi-line and heat pump system internal and external machines, refrigerant Automatic flow regulation, so that the system in the best operating conditions, fast cooling or heating, accurate control and energy saving.



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R410A、R32、R290、R22、R407C、R454B、R314a、R507等
◆ 容量范围 Capacity Range	40kW及以下（R410A名义容量） 40kW MAX（R410A Nominal capacity）
◆ 冷媒温度 Refrigerant Temperature	-40℃~+85℃（通电率50%以下） -40℃~+85℃（duty ratio 50% MAX）
◆ 环境温度 Ambient Temperature	-30℃~+60℃（通电率50%以下） -30℃~+60℃（duty ratio 50% MAX）
◆ 环境湿度 Ambient Humidity	≤95%RH
◆ 介质流向 Flow Direction	正反皆可 Bi-flow
◆ 全开脉冲 Full Stroke Pulses	500P
◆ 开阀脉冲 On-valve Pulses	32±20P
◆ 安装方式 Installation Method	以阀体及线圈的断面中心线为轴，并且将线圈朝上，在垂直±15°以内 Take the center line of the valve body and coil section as the axis, and turn the coil up, within the vertical ±15°

电气参数 ELECTRICAL PARAMETER		
◆ 额定电压 Rated Voltage	DC12V±10%，矩形波 DC12V±10%，Rectangular wave	
◆ 驱动方式 Actuating Mechanism	直动式4相8拍永磁步进电机 4-phase 8-beat permanent magnet stepper motor,direct-acting	
◆ 励磁方式 Recommended Drive Condition	1~2相励磁，单极驱动 1~2 phase excitation,Unipolar drive	
◆ 励磁速度 Excitation Rate	Φ1.0~Φ2.4mm :30~90PPS Φ2.5~Φ3.2mm :30~40PPS	
◆ 额定电阻 Rated Resistance	46±3.7Ω/相（20℃） 46±3.7Ω/ phase(20℃)	
◆ 额定电流 Rated Current	0.26A/相（20℃） 0.26A/ phase(20℃)	
◆ 绝缘等级 Insulation Class	E级 Class E	
◆ 防护等级 Protection Class	IP67	

技术参数

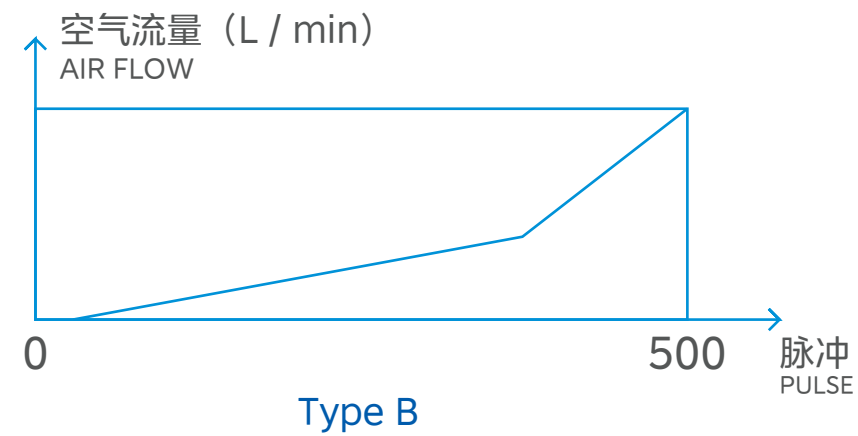
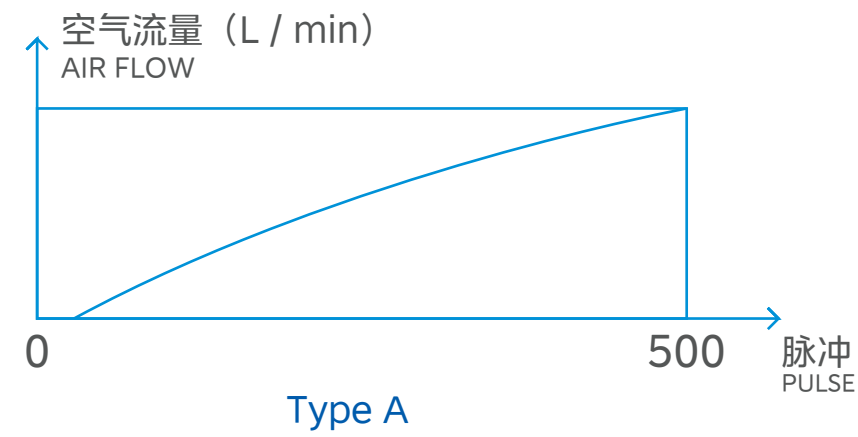
Technical Parameter

系列 Production Series	口径 Port Diameter	名义容量(kW) Nominal Capacity						最大工作压力 MAX Operating Pressure	最大动作压差 Max Operating Pressure Difference	逆向开阀压差 Converse On-valve Operating Pressure Difference
-	mm	R410A	R32	R290	R22	R134a	R407C	MPa	MPa	MPa
DPFW*-13*	1.3	6.1	9.2	5.6	5.2	4.1	5.4	4.9	3.5	≥2.5
DPFW*-16*	1.7	10.4	15.5	9.4	8.9	6.9	9.2			
DPFW*-18*	1.8	12.4	18.4	11.1	10.6	8.3	10.9			
DPFW*-20*	2.0	15.3	22.8	13.7	13.1	10.2	13.5			
DPFW*-24*	2.4	19.4	28.8	17.4	16.6	12.9	17.1			
DPFW*-32*	3.2	35.5	52.6	31.7	27.8	23.7	31.3			≥2.1

注：额定工况：冷凝温度38℃；蒸发温度5℃；过冷度0℃；过热度0℃。
Rated working condition: condensing temp. 38 °C, evaporating temp. 5 °C, sub-cooled 0 °C, overheat 0 °C.

流量特点

Flow Characteristics



外形尺寸

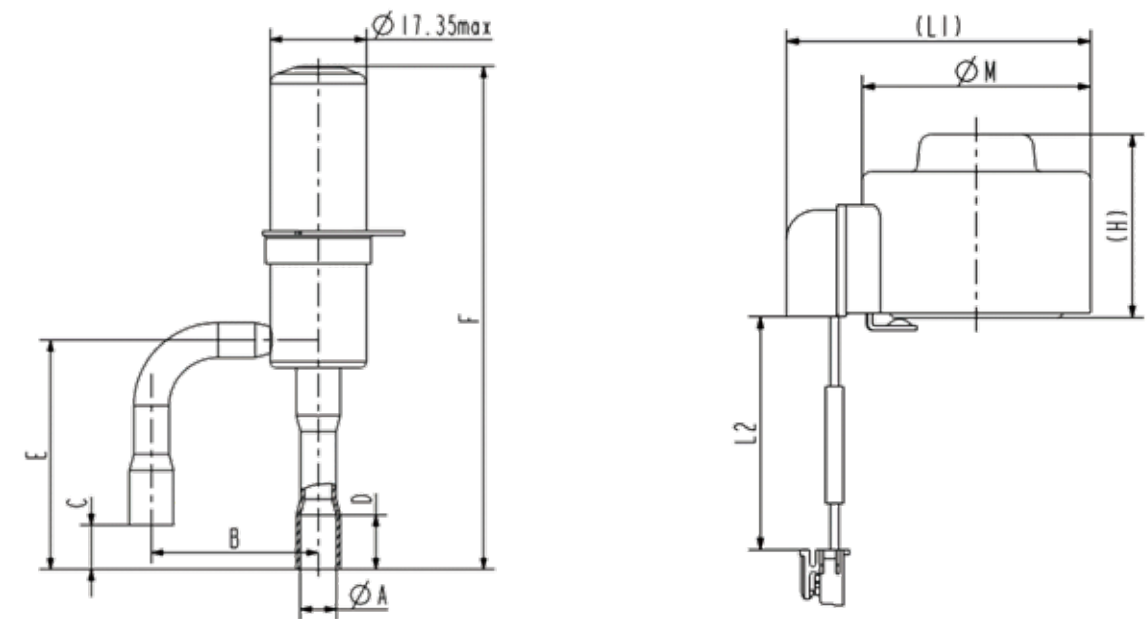
External Dimension

阀体

VALVE BODY

线圈

COIL



阀系列 Valve Shape	阀体基本尺寸 (mm) Basic Size of Valve					
	A	B	C	D	E	F
DPFW*-13~32*	6.5	30	8	10	60	110

线圈系列 Coil Series	线圈基本尺寸 (mm) Basic Size of Coil				连接器型号 Housing Type
	H	M	L1	L2	
DPFW*-13~32*	30.5	38.5	51	750	XHP-5
					XHP-6

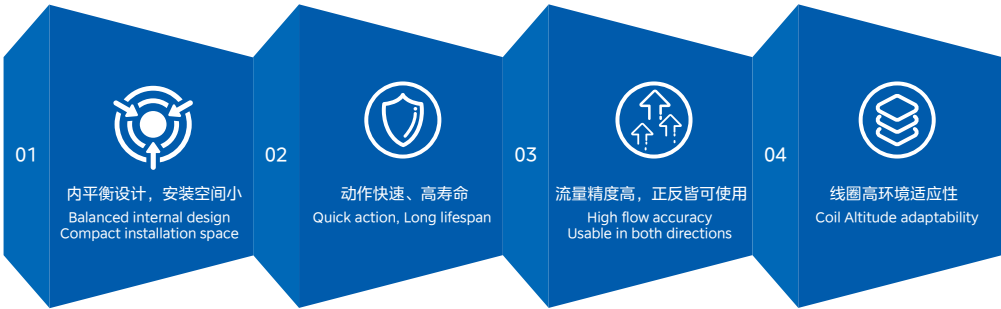
* 线圈导线长度及连接器型号可定制 Wire length and housing type can be customized



产品应用与特点
Product application and features

N系列电子膨胀阀主要应用于多联机外机、模块机、单元机、热泵热水机以及商用制冷等系统中，实现制冷剂流量的自动调节，使系统在最佳工况下运行，快速制冷或制热，精确控温和节能。

N series electronic expansion valve is mainly used in systems such as VRF outdoor units, modular units, unitary units, heat pump water heaters, and commercial refrigeration, to realize the automatic adjustment of refrigerant flow. Make the system run in the best condition, fast cooling or heating, accurate control and energysaving.



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R410A、R32、R290、R22、R407C、R454B、R314a、R507等
◆ 容量范围 Capacity Range	40kW~120kW（R410A名义容量） 40kW~120kW（R410A Nominal capacity）
◆ 冷媒温度 Refrigerant Temperature	-40℃~+85℃（通电率50%以下） -40℃~+85℃（duty ratio 50% MAX）
◆ 环境温度 Ambient Temperature	-30℃~+60℃（通电率50%以下） -30℃~+60℃（duty ratio 50% MAX）
◆ 环境湿度 Ambient Humidity	≤95%RH
◆ 介质流向 Flow Direction	正反皆可 Bi-flow
◆ 全开脉冲 Full Stroke Pulses	500P
◆ 开阀脉冲 On-valve Pulses	32±20P
◆ 安装方式 Installation Method	以阀体及线圈的断面中心线为轴，并且将线圈朝上，在垂直±15°以内 Take the center line of the valve body and coil section as the axis, and turn the coil up, within the vertical ±15°

电气参数 ELECTRICAL PARAMETER		
◆ 额定电压 Rated Voltage	DC12V±10%，矩形波 DC12V±10%，Rectangular wave	
◆ 驱动方式 Actuating Mechanism	直动式4相8拍永磁步进电机 4-phase 8-beat permanent magnet stepper motor,direct-acting	
◆ 励磁方式 Recommended Drive Condition	1~2相励磁，单极驱动 1~2 phase excitation,Unipolar drive	
◆ 励磁速度 Excitation Rate	φ3.5~φ6.5mm :30~90PPS	
◆ 额定电阻 Rated Resistance	46±3.7Ω/ 相 (20℃) 46±3.7Ω/ phase(20℃)	
◆ 额定电流 Rated Current	0.26A/ 相 (20℃) 0.26A/ phase(20℃)	
◆ 绝缘等级 Insulation Class	E级 Class E	
◆ 防护等级 Protection Class	IP67	

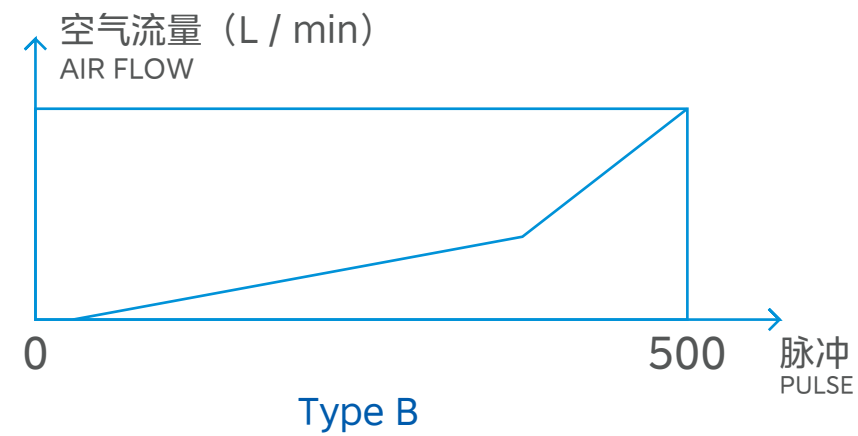
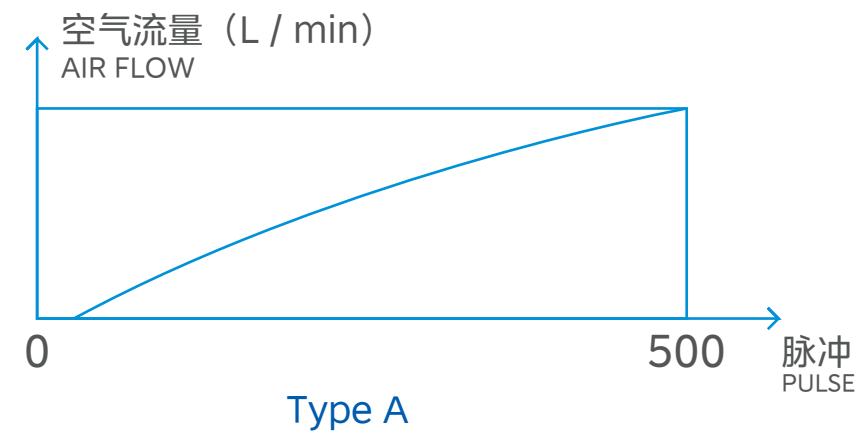
技术参数
Technical Parameter

系列 Production Series	口径 Port Diameter	名义容量(kW) Nominal Capacity					最大工作压力 MAX Operating Pressure	最大动作压差 Max Operating Pressure Difference	逆向开阀压差 Converse On-valve Operating Pressure Difference
		R410A	R32	R290	R134a	R407C	MPa	MPa	MPa
DPFN*-35*	3.5	42.6	62.8	37.9	28.7	28.6	4.9	3.5	3.5
DPFN*-40*	4.0	59.0	87.5	52.8	39.3	51.9			
DPFN*-45*	4.5	79.5	117.0	70.6	53.0	70.0			
DPFN*-50*	5.0	85.2	126.4	76.5	57.1	75.6			
DPFN*-55*	5.5	91.5	135.5	81.7	61.0	80.6			
DPFN*-60*	6.0	102.3	150.2	91.2	68.0	89.8			
DPFN*-65*	6.5	111.8	164.8	99.4	74.5	98.4			

注：额定工况：冷凝温度38℃；蒸发温度5℃；过冷度0℃；过热度0℃。
Rated working condition: condensing temp. 38 °C, evaporating temp. 5 °C, sub-cooled 0 °C, overheat 0 °C.

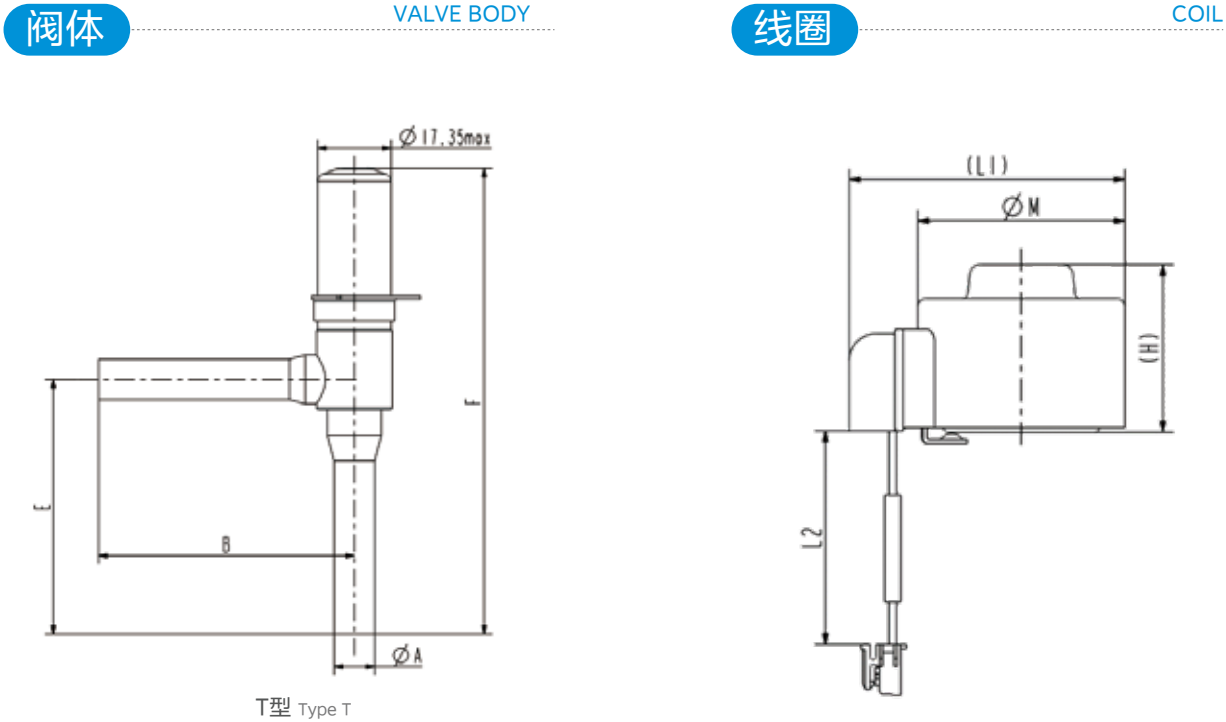
流量特点

Flow Characteristics



外形尺寸

External Dimension



外形 Shape	阀系列 Valve Series	阀体基本尺寸 (mm) Basic Size of Valve			
		A	B	E	F
T型 Type T	DPFN*-35~65*	9.52/12.7/15.88	60	60	110

线圈系列 Coil Series	线圈基本尺寸 (mm) Basic Size of Coil				连接器型号 Housing Type
	H	M	L1	L2	
DPFN*-35~65*	30.5	38.5	51	1600	XHP-5
					XHP-6

* 线圈导线长度及连接器型号可定制 Wire length and housing type can be customized



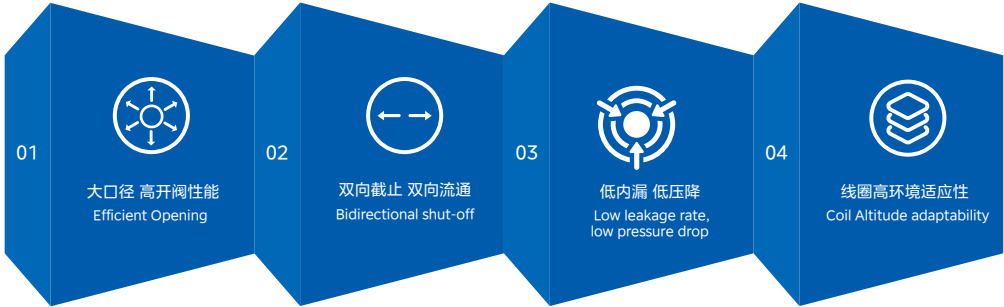
L系列电子膨胀阀
L SERIES EEV



产品应用与特点
Product application and features

L系列电子膨胀阀主要应用于大型商用空调或易燃冷媒系统中，实现制冷剂调节或双向截断，使系统在最佳工况下安全运行。

L series electronic expansion valve is mainly used in large commercial air conditioning systems or flammable refrigerant systems to regulate refrigerant flow or shut off flow in both directions, ensuring safe operation of the system under optimal conditions.



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R410A、R32、R290、R407C、R454B、R314a、R507等
◆ 容量范围 Capacity Range	120kW及以上（R410A名义容量） 120kW MIN（R410A Nominal capacity）
◆ 冷媒温度 Refrigerant Temperature	-40℃~+85℃（通电率50%以下） -40℃~+85℃（duty ratio 50% MAX）
◆ 环境温度 Ambient Temperature	-30℃~+60℃（通电率50%以下） -30℃~+60℃（duty ratio 50% MAX）
◆ 环境湿度 Ambient Humidity	≤95%RH
◆ 介质流向 Flow Direction	正反皆可 Bi-flow
◆ 全开脉冲 Full Stroke Pulses	500P
◆ 安装方式 Installation Method	以阀体及线圈的断面中心线为轴，并且将线圈朝上，在垂直±15°以内 Take the center line of the valve body and coil section as the axis, and turn the coil up, within the vertical ±15°

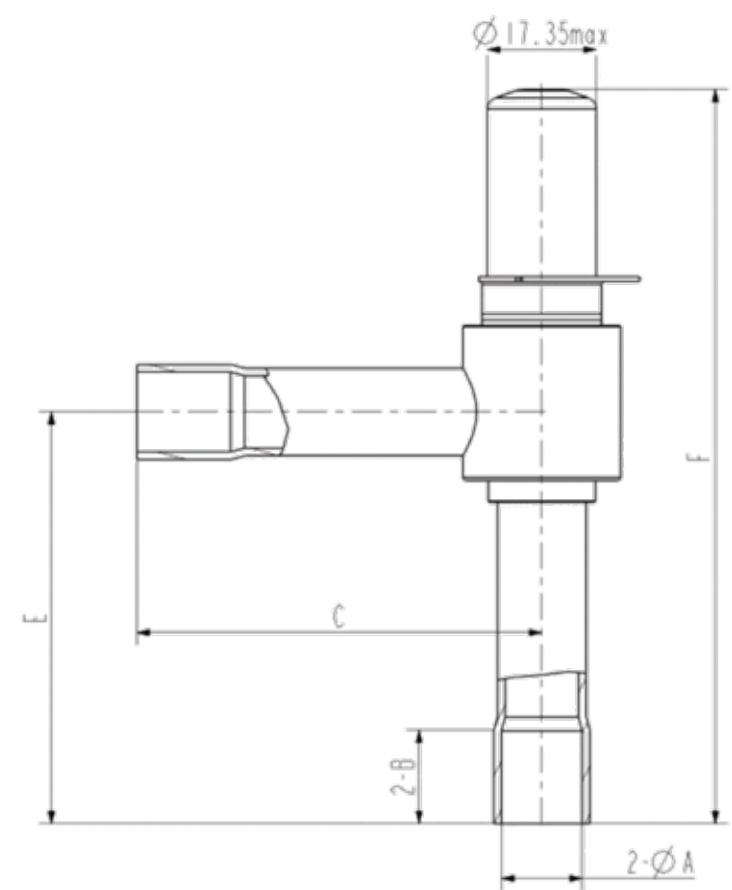
电气参数 ELECTRICAL PARAMETER		
◆ 额定电压 Rated Voltage	DC12V±10%，矩形波 DC12V±10%,Rectangular wave	
◆ 驱动方式 Actuating Mechanism	直动式4相8拍永磁步进电机 4-phase 8-beat permanent magnet stepper motor,direct-acting	
◆ 励磁方式 Recommended Drive Condition	1~2相励磁，单极驱动 1~2 phase excitation,Unipolar drive	
◆ 励磁速度 Excitation Rate	30 ~ 40PPS	
◆ 额定电阻 Rated Resistance	46±3.7Ω/相(20℃) 46±3.7Ω/ phase(20℃)	
◆ 额定电流 Rated Current	0.26A/相(20℃) 0.26A/ phase(20℃)	
◆ 绝缘等级 Insulation Class	E级 Class E	
◆ 防护等级 Protection Class	IP67	

技术参数
Technical Parameter

系列 Production Series	口径 Port Diameter	Cv	内漏 Leakage Rate	最大工作压力 MAX Operating Pressure	最大动作压差 Max Operating Pressure Difference
-	mm	/	ml/min	Mpa	Mpa
DPFL*-80*	8	2	< 300	4.3	3.5
DPFL*-100*	10	3	< 300		
DPFL*-130*	13	4	< 300		

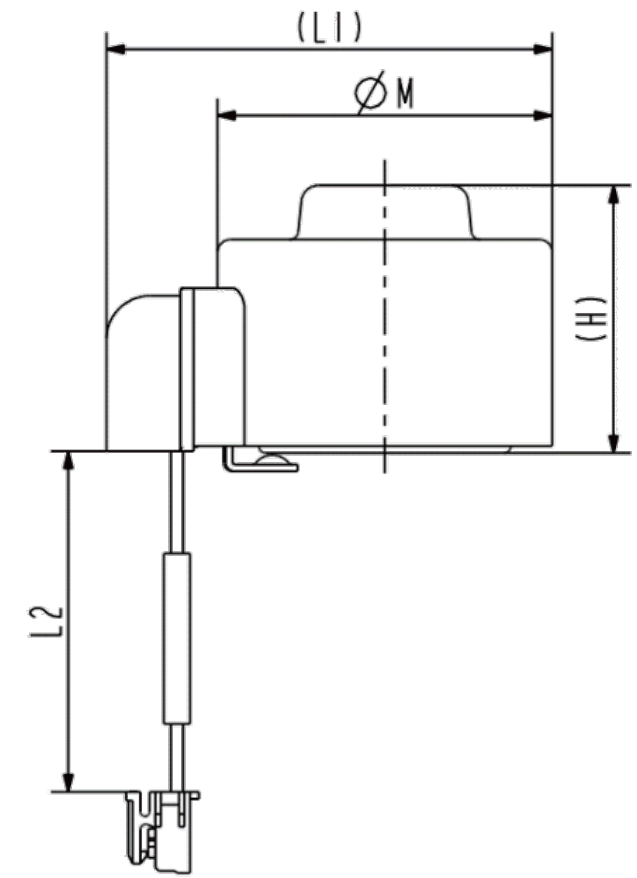
外形尺寸
External Dimension

阀体 VALVE BODY



阀系列 Valve Shape	阀体基本尺寸 (mm) Basic Size of Valve				
	A	B	C	E	F
DPFL*-80*	16	12	55	55	106
DPFL*-100*	16	12	55	55	106
DPFL*-130*	16	12	55	55	106

线圈 COIL

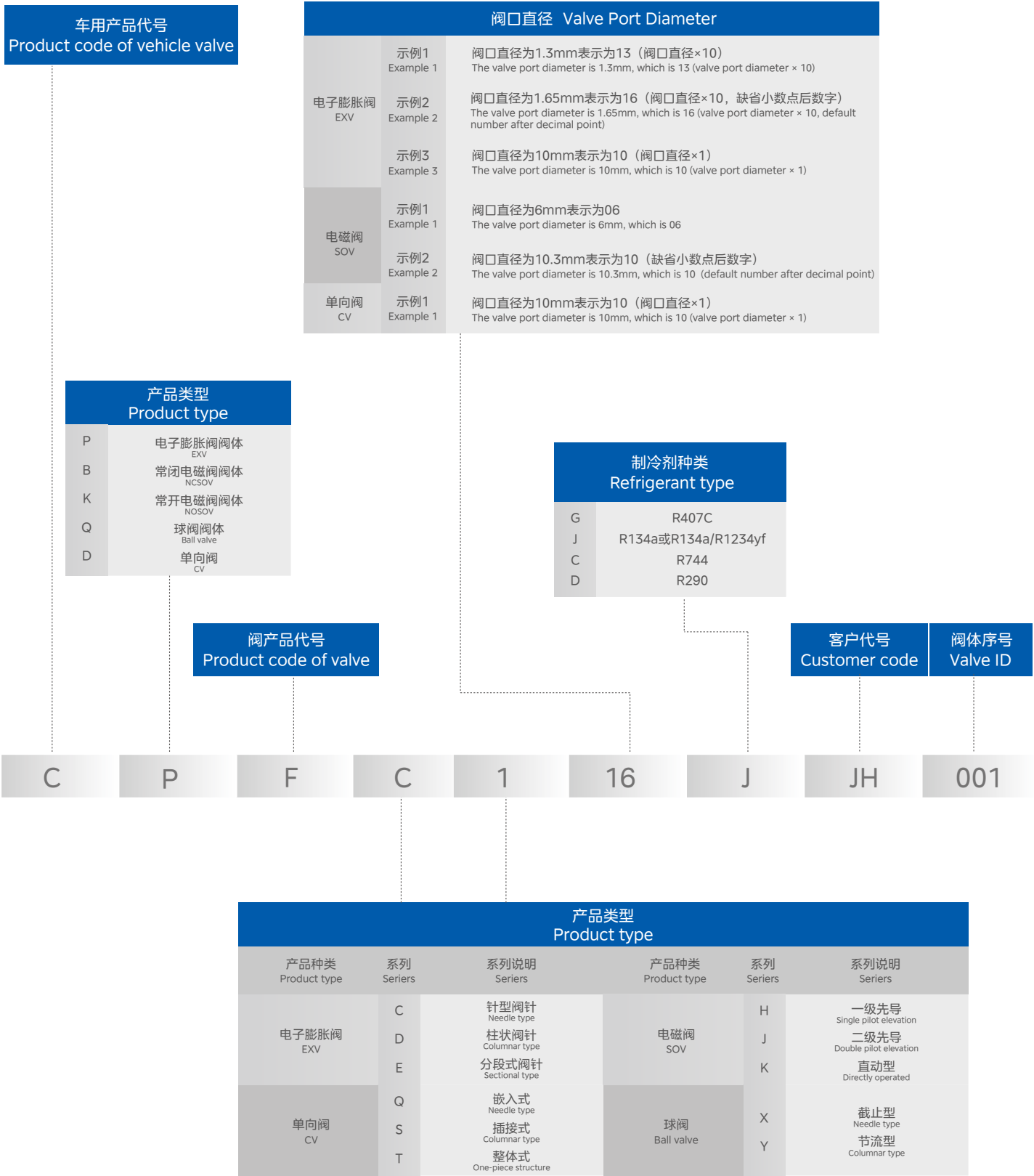


线圈系列 Coil Series	线圈基本尺寸 (mm) Basic Size of Coil				连接器型号 Housing Type
	H	M	L1	L2	
DPFN*-80~130*	30.5	38.5	51	1500	XHP-5
					XHP-6

* 线圈导线长度及连接器型号可定制 Wire length and housing type can be customized

车用阀命名规则

Vehicle Valve Naming Scheme





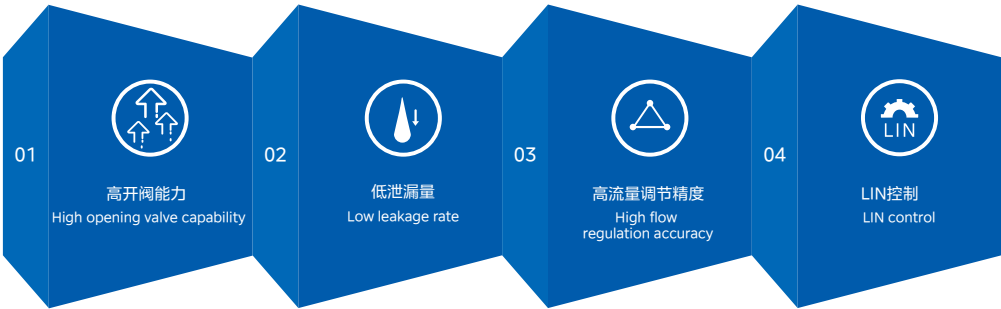
电气参数 ELECTRICAL PARAMETER		
◆ 额定电压	Rated Voltage	DC12V
◆ 驱动方式	Drive Mode	2相8拍永磁步进电机 2-phase 8-beat permanent magnet stepper motor
◆ 励磁方式	Excitation Method	1~2相励磁，双极驱动 1-2 phase excitation, bipolar drive
◆ 励磁速度	Excitation Speed	≤200PPS
◆ 额定电阻	Rated Resistance	23±2Ω/相(20℃)
◆ 额定电流	Rated Current	0.155A(20℃)
◆ 绝缘等级	Insulation Class	E级 Grade E
◆ 防护等级	Protection Class	IP6K9K



产品应用与特点
Product application and features

C1系列电子膨胀阀主要应用于车用热管理系统中，实现制冷剂流量的自动调节，使系统在最佳工况下运行，快速制冷或制热，精确控温和节能。

The C1 series electronic expansion valve is mainly used in automotive thermal management systems to achieve automatic adjustment of refrigerant flow, enabling the system to operate under optimal conditions, quickly cool or heat, accurately control temperature, and save energy.



技术参数

Technical Parameter

系列 Small Series	口径 Caliber	名义容量(kW) Nominal Capacity		最大工作压力 Maximum Work Pressure	最大工作压差 Maximum Working Pressure Difference
-	mm	R134a/ R1234yf	R290	MPa	MPa
CPFC1-12J*	1.2	3.5	4.6	3.5	2.5
CPFC1-16J*	1.65	6.9	9.0		
CPFC1-18J*	1.8	8.3	10.8		
CPFC1-20J*	2.0	10.2	13.3		
CPFC1-24J*	2.4	12.9	16.8		
CPFC1-32J*	3.2	23.7	30.8		

注：额定工况：冷凝温度38℃；蒸发温度5℃；过冷度0℃；过热度0℃。
Note: Rated operating conditions: condensing temperature 38 °C; Evaporation temperature 5 °C; Subcooling degree 0 °C; Overheating degree 0 °C.

基本规格 BASIC SPECIFICATION	
◆ 适用冷媒	Suitable Refrigerant R134a/R1234yf/R290
◆ 口径范围	Caliber Range 1.2mm~3.2mm
◆ 冷媒温度	Refrigerant Temperature -40℃~+85℃（通电率50%以下 Power on rate below 50%）
◆ 环境温度	Ambient Temperature -40℃~+125℃（通电率50%以下 Power on rate below 50%）
◆ 环境湿度	Ambient Humidity ≤95%RH
◆ 介质流向	Flow Direction 正反皆可 Bi-flow
◆ 全开脉冲	Full Stroke Pulse 576P
◆ 开阀脉冲	On-valve Pulses 38±15P
◆ 安装方式	Installation Method 阀体通过自身螺纹固定，扭矩15~20N·m，线圈与阀体装配后通过螺钉组件固定，扭矩3~6N·m The valve body is fixed by its own threads with a torque of 15~20N·m, and the coil and valve body are fixed by screw components after assembly with a torque of 3~6N·m

电气参数 ELECTRICAL PARAMETER		
◆ 额定电压	Rated Voltage	DC12V
◆ 驱动方式	Drive Mode	2相8拍永磁步进电机 2-phase 8-beat permanent magnet stepper motor
◆ 励磁方式	Excitation Method	1~2相励磁, 双极驱动 1-2 phase excitation, bipolar drive
◆ 励磁速度	Excitation Speed	≤200PPS
◆ 额定电阻	Rated Resistance	23±2Ω/相(20℃)
◆ 额定电流	Rated Current	0.52A/相(20℃)
◆ 绝缘等级	Insulation Class	E级 Grade E
◆ 防护等级	Protection Class	IP6K9K

技术参数

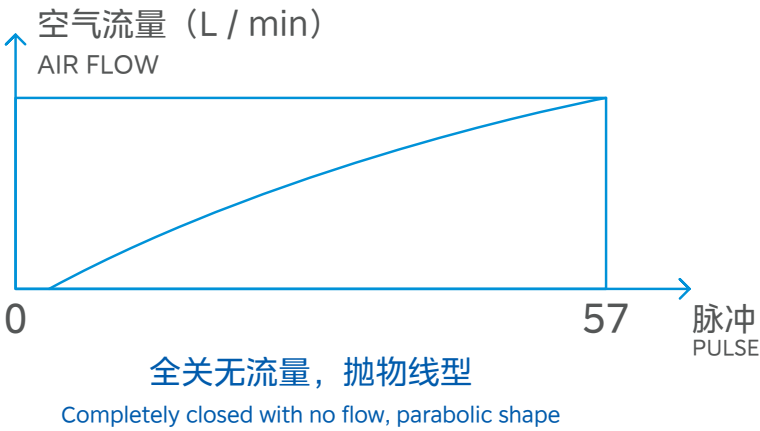
Technical Parameter

系列 Series	口径 Caliber	名义容量(kW) Nominal Capacity		最大工作压力 Maximum Work Pressure	最大工作压差 Maximum Working Pressure Difference
		R134a/ R1234yf	R290		
CPFD1-45*	4.5	4.1	5.3	3.5	2.5
CPFD1-10*	10.0	7.2	9.4		
CPFD1-16*	16.0	19.6	25.5		

注：额定工况：冷凝温度38℃；蒸发温度5℃；过冷度0℃；过热度0℃。
Note: Rated operating conditions: condensing temperature 38 °C; Evaporation temperature 5 °C; Subcooling degree 0 °C; Overheating degree 0 °C.

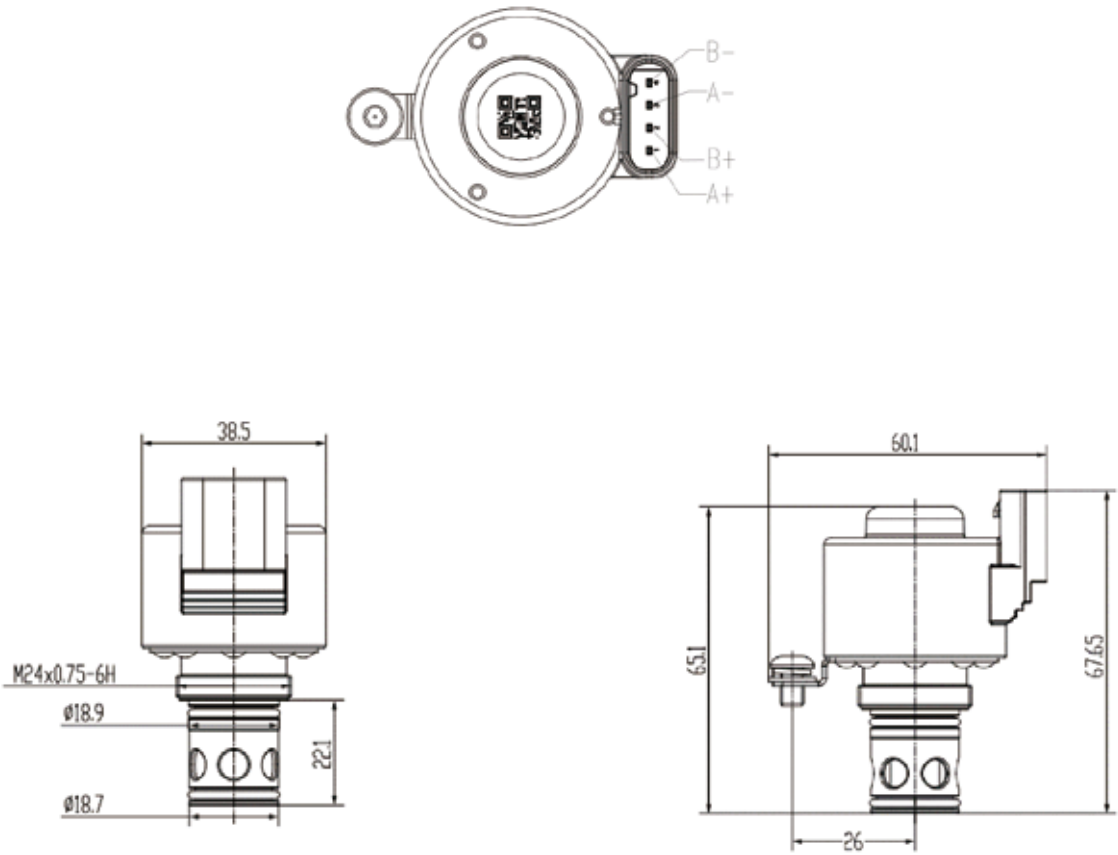
流量特点

Traffic Characteristics



外形尺寸

External Dimension





E1系列电子膨胀阀

E1 SERIES ELECTRONIC EXPANSION VALVE

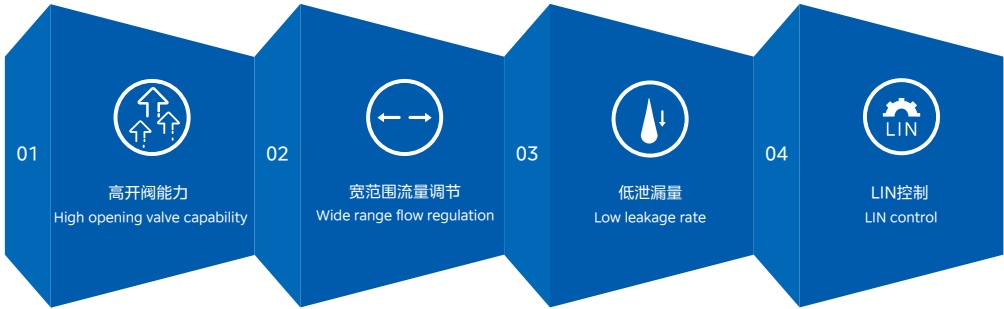


产品应用与特点

Product application and features

E1系列电子膨胀阀主要应用于车用热管理系统中，实现小开度下精确调节流量，全开时实现低压降，满足全流通功能，同时具备膨胀和全通功能，简化热管理系统。

The E1 series electronic expansion valve is mainly used in automotive thermal management systems to achieve precise flow regulation under small opening degrees, achieve low pressure drop when fully opened, and meet full flow functions. It also has expansion and full pass functions, simplifying the thermal management system.



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R134a/R1234yf/R290
◆ 口径范围 Caliber Range	10mm~16mm
◆ 冷媒温度 Refrigerant Temperature	-40℃~+85℃（通电率50%以下 Power on rate below 50%）
◆ 环境温度 Ambient Temperature	-40℃~+125℃（通电率50%以下 Power on rate below 50%）
◆ 环境湿度 Ambient Humidity	≤95%RH
◆ 介质流向 Flow Direction	正反皆可 Bi-flow
◆ 全开脉冲 Full Stroke Pulse	768P
◆ 开阀脉冲 On-valve Pulses	38±15P
◆ 安装方式 Installation Method	阀体通过自身螺纹固定，扭矩15~20N·m，线圈与阀体装配后通过螺钉组件固定，扭矩3~6N·m The valve body is fixed by its own threads with a torque of 15~20N·m, and the coil and valve body are fixed by screw components after assembly with a torque of 3~6N·m

电气参数 ELECTRICAL PARAMETER		
◆ 额定电压 Rated Voltage	DC12V	
◆ 驱动方式 Drive Mode	直动式2相8拍永磁步进电机 2-phase 8-beat permanent magnet stepper motor	
◆ 励磁方式 Excitation Method	1~2相励磁，双极驱动 1-2 phase excitation, bipolar drive	
◆ 励磁速度 Excitation Speed	≤200PPS	
◆ 额定电阻 Rated Resistance	23±2Ω/相(20℃)	
◆ 额定电流 Rated Current	0.155A(20℃)	
◆ 绝缘等级 Insulation Class	E级 Grade E	
◆ 防护等级 Protection Class	IP6K9K	

技术参数

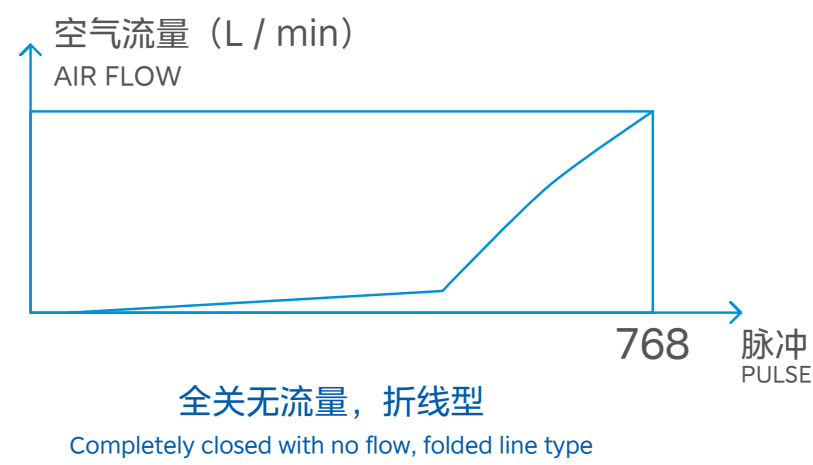
Technical Parameter

系列 Series	口径 Caliber	名义容量(kW) Nominal Capacity		最大工作压力 Maximum Work Pressure	最大工作压差 Maximum Working Pressure Difference
-	mm	R134a/ R1234yf	R290	MPa	MPa
CPFE1-12J*	12	12.2	15.9	3.5	2.5
CPFE1-16J*	16	19.6	25.5		

注：额定工况：冷凝温度38℃；蒸发温度5℃；过冷度0℃；过热度0℃。
Note: Rated operating conditions: condensing temperature 38℃; Evaporation temperature 5℃; Subcooling degree 0℃; Overheating degree 0℃.

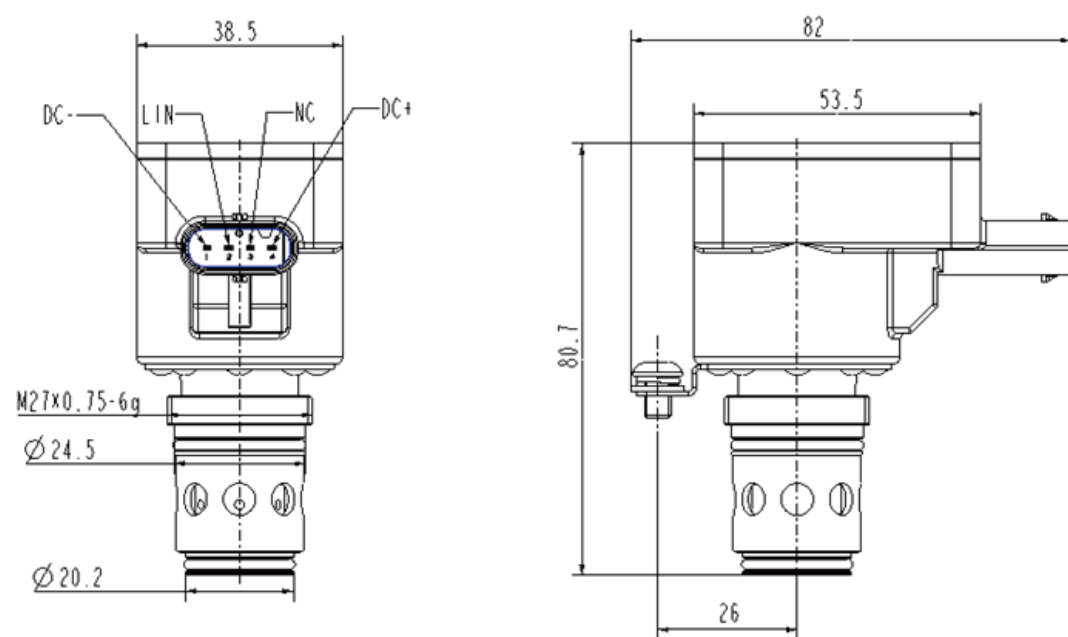
流量特点

Traffic Characteristics



外形尺寸

External Dimension

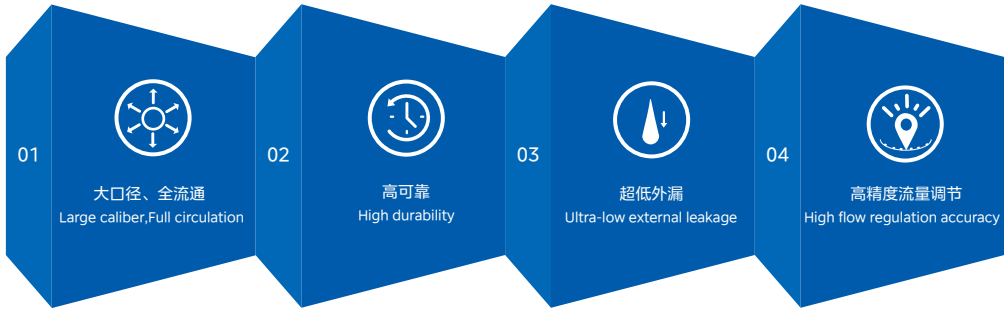


产品应用与特点

Product application and features

D2系列电子膨胀阀主要应用于车用热管理R744系统中，具备流量分配、全通功能，满足系统不同工况使用要求。

D2 series electronic expansion valve is mainly used in vehicle thermal management R744 system, with flow distribution, full pass functions, to meet the requirements of different operating conditions of the system.



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R744
◆ 口径范围 Caliber Range	7mm~10mm
◆ 冷媒温度 Refrigerant Temperature	-40℃~+165℃ (通电率50%以下 Power on rate below 50%)
◆ 环境温度 Ambient Temperature	-40℃~+140℃ (通电率50%以下 Power on rate below 50%)
◆ 环境湿度 Ambient Humidity	≤95%RH
◆ 介质流向 Flow Direction	正反皆可 Bi-flow
◆ 全开脉冲 Full Stroke Pulse	860P
◆ 开阀脉冲 On-valve Pulses	32±20P

电气参数 ELECTRICAL PARAMETER		
◆ 额定电压	Rated Voltage	DC12V±10%
◆ 驱动方式	Drive Mode	3相永磁步进电机 3-phase permanent magnet stepper motor
◆ 励磁方式	Excitation Method	两相励磁 2 phase excitation
◆ 励磁速度	Excitation Speed	30~100PPS
◆ 额定电阻	Rated Resistance	8.2±5%Ω/相(20℃)
◆ 额定电流	Rated Current	0.73A/相(20℃)
◆ 绝缘等级	Insulation Class	H级 Grade H
◆ 防护等级	Protection Class	IP6K9K

技术参数

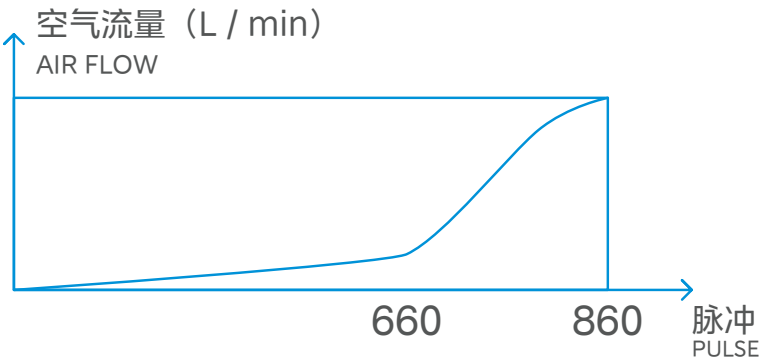
Technical Parameter

系列 Series	口径 Caliber	名义容量(kW) Nominal Capacity	最大工作压力 Maximum Work Pressure	最大工作压差 Maximum Working Pressure Difference
-	mm	R744	MPa	MPa
CPFD2-07*	7.0	4.1	17	10.5
CPFD2-10*	10.0	7.2		

注：额定工况：冷凝温度38℃；蒸发温度5℃；过冷度0℃；过热度0℃。
Note: Rated operating conditions: condensing temperature 38 °C; Evaporation temperature 5 °C; Subcooling degree 0 °C; Overheating degree 0 °C.

流量特点

Traffic Characteristics

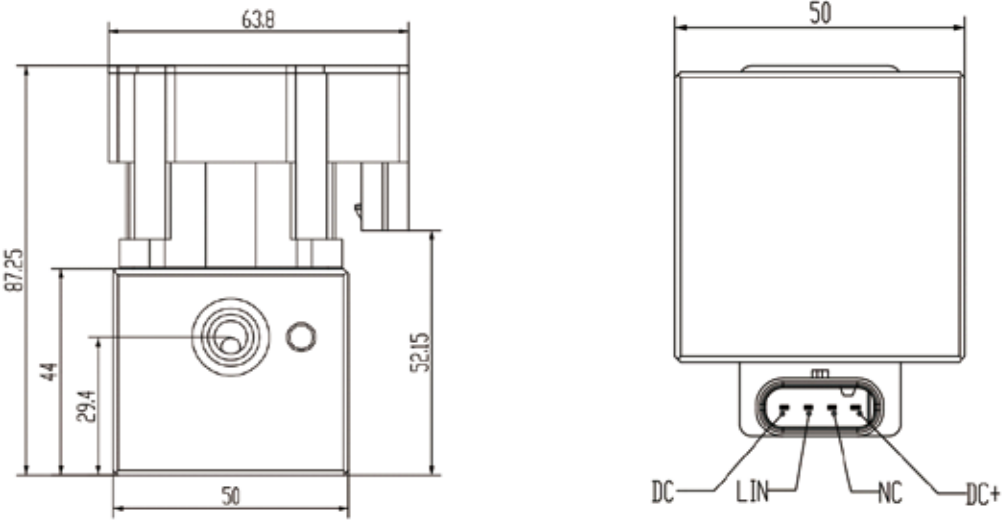


全关无流量，前段流量调节，后段全流通

Upstream flow regulation and downstream full circulation

外形尺寸

External Dimension





H1系列电磁阀

H1 SERIES SOLENOID VALVES

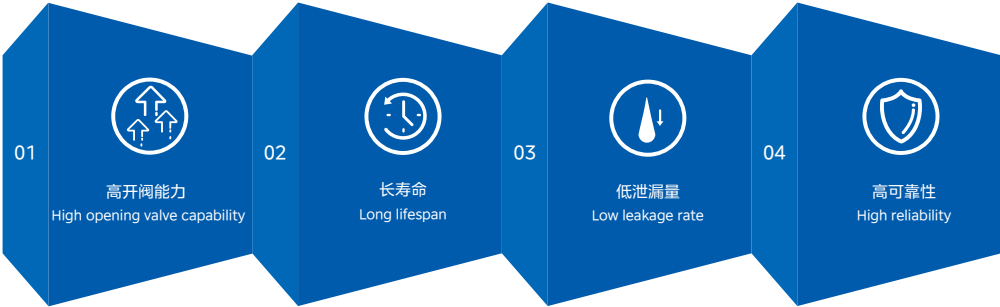


产品应用与特点

Product application and features

H1系列电磁阀主要应用于车用热管理系统中，实现制冷剂流路的自动通断，满足系统制冷或制热不同运行模式的需求。

The H1 series solenoid valve is mainly used in automotive thermal management systems to achieve automatic on-off of refrigerant flow paths, meeting the needs of different operating modes of system cooling or heating.



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R134a/R1234yf/R290
◆ 口径范围 Caliber Range	6mm~12mm
◆ 冷媒温度 Refrigerant Temperature	-40℃~+125℃
◆ 环境温度 Ambient Temperature	-40℃~+125℃
◆ 环境湿度 Ambient Humidity	≤98%RH
◆ 工作压力 Working Pressure	3.5MPa
◆ 工作压差 Operating differential pressure	≥2.5MPa (DC9V, 20℃)
◆ 内泄漏量 Internal leakage flowrate	≤30ml/min
◆ 安装方式 Installation Method	阀体通过自身螺纹固定，线圈与阀体装配后通过螺钉组件固定。 The valve body is fixed by its own threads, and the coil and valve body are fixed by screw components after assembly.

电气参数 ELECTRICAL PARAMETER		
◆ 额定电压 Rated Voltage		DC12V
◆ 电压范围 Voltage Range		DC9V~DC16V
◆ 额定电阻 Rated Resistance		14.8Ω±5%Ω(20℃)
◆ 额定电流 Rated Current		≤1A (20℃, DV12V)
◆ 绝缘电阻 Insulation Resistance		≥100MΩ
◆ 绝缘等级 Insulation Class		H级 Grade H
◆ 防护等级 Protection Class		IP6K9K

技术参数

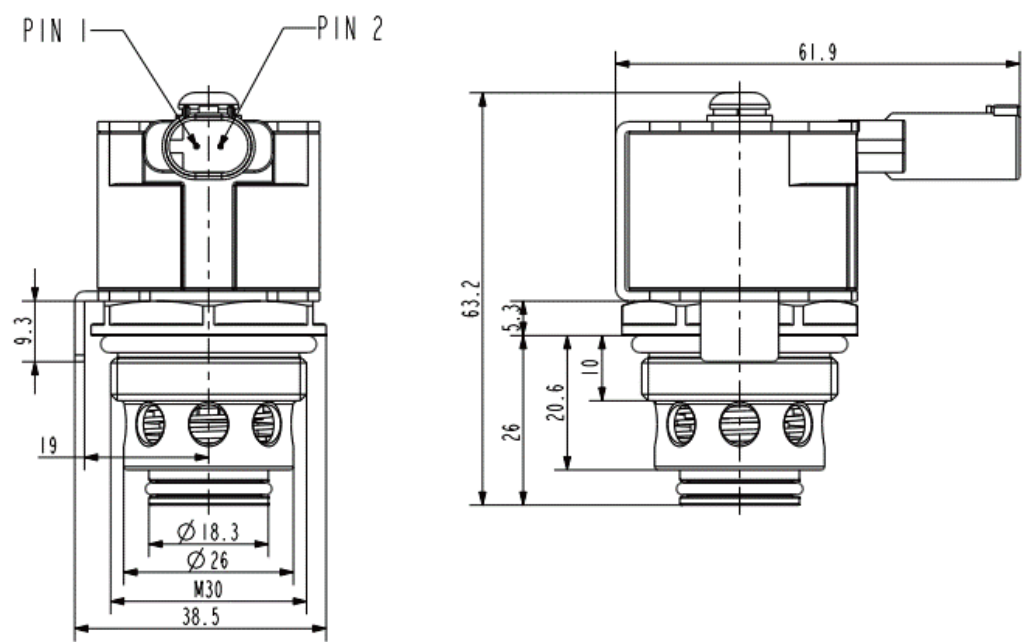
Technical Parameter

系列 Series	口径 Bore	名义容量(kW) Theoretical Capacity		最大工作压力 Maximum Work Pressure	最大工作压差 Maximum Working Pressure Difference	动作型式 Operating Mode
		R134a/ R1234yf	R290	MPa	MPa	
CBFH1-06*	6.0	10.0	12.0	3.5	2.5	常闭式电磁阀 Normally closed solenoid valve
CBFH1-08*	8.0	14.8	17.8			
CBFH1-10*	10.0	27.9	33.6			
CBFH1-12*	12.0	34.8	41.9			
CKFH1-06*	6.0	10.0	12.0			常开式电磁阀 Normally open solenoid valve
CKFH1-08*	8.0	14.8	17.8			
CKFH1-10*	10.0	27.9	33.6			
CKFH1-12*	12.0	34.8	41.9			

注：系统工况：冷凝温度25℃；蒸发温度-10℃。
Note: System operating conditions: condensation temperature 25 °C; Evaporation temperature -10 °C.

外形尺寸
External Dimension

The series of CBFH1 CBFH1系列

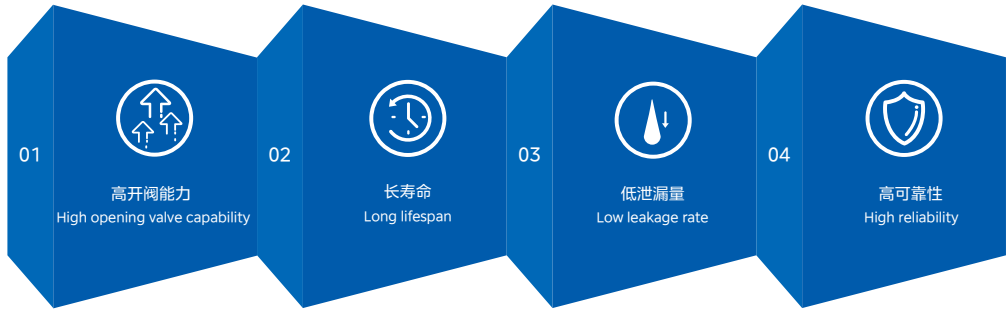
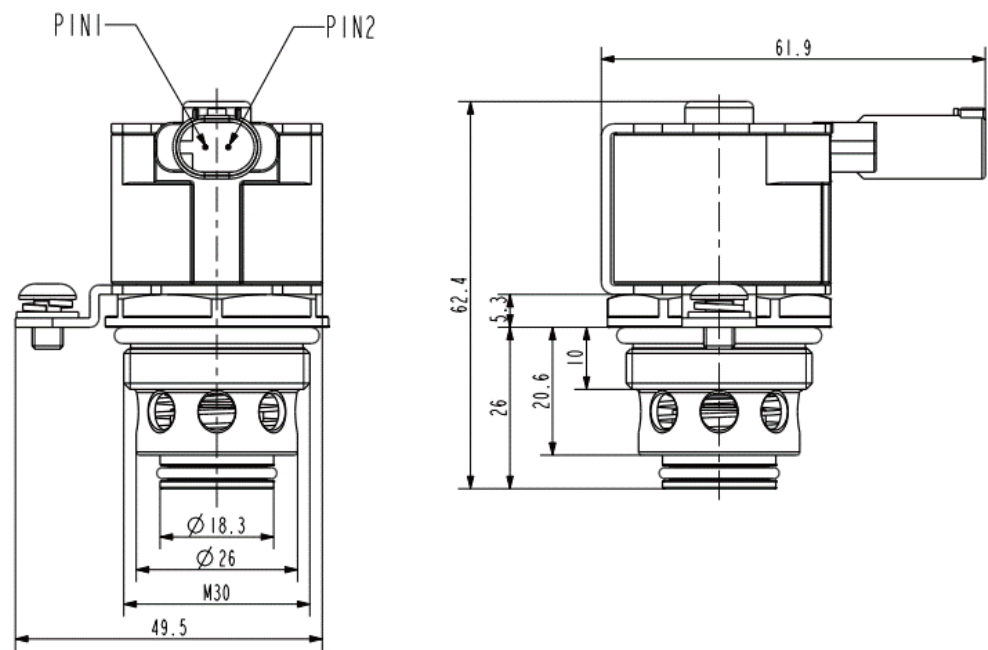


产品应用与特点
Product application and features

H2系列电磁阀主要应用于车用热管理系统中，实现制冷剂流路的自动通断，满足系统制冷或制热不同运行模式的需求。

The H2 series solenoid valve is mainly used in automotive thermal management systems to achieve automatic on-off of refrigerant flow paths, meeting the needs of different operating modes of system cooling or heating.

The series of CKFH1 CKFH1系列



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R134a/R1234yf /R290
◆ 口径范围 Caliber Range	6mm~12mm
◆ 冷媒温度 Refrigerant Temperature	-40℃~+125℃
◆ 环境温度 Ambient Temperature	-40℃~+125℃
◆ 环境湿度 Ambient Humidity	≤98%RH
◆ 工作压力 Working Pressure	3.5MPa
◆ 工作压差 Operating differential pressure	≥2.5MPa (DC18V, 20℃)
◆ 内泄漏量 Internal leakage flowrate	≤30ml/min
◆ 安装方式 Installation Method	阀岛式结构，阀体通过进、出口与管接头连接固定，线圈与阀体装配后通过螺钉组合件固定。 Valve island structure, the valve body is connected and fixed to the pipe joints through the inlet and outlet, and the coil and valve body are assembled and fixed by screw assemblies.

电气参数 ELECTRICAL PARAMETER		
◆ 额定电压	Rated Voltage	DC24V
◆ 电压范围	Voltage Range	DC18V~DC24V
◆ 额定电阻	Rated Resistance	51Ω±5%Ω(20℃)
◆ 额定电流	Rated Current	≤0.5A (20℃, DV24V)
◆ 绝缘电阻	Insulation Resistance	≥100MΩ
◆ 绝缘等级	Insulation Class	H级 Grade H
◆ 防护等级	Protection Class	IP6K9K

技术参数

Technical Parameter

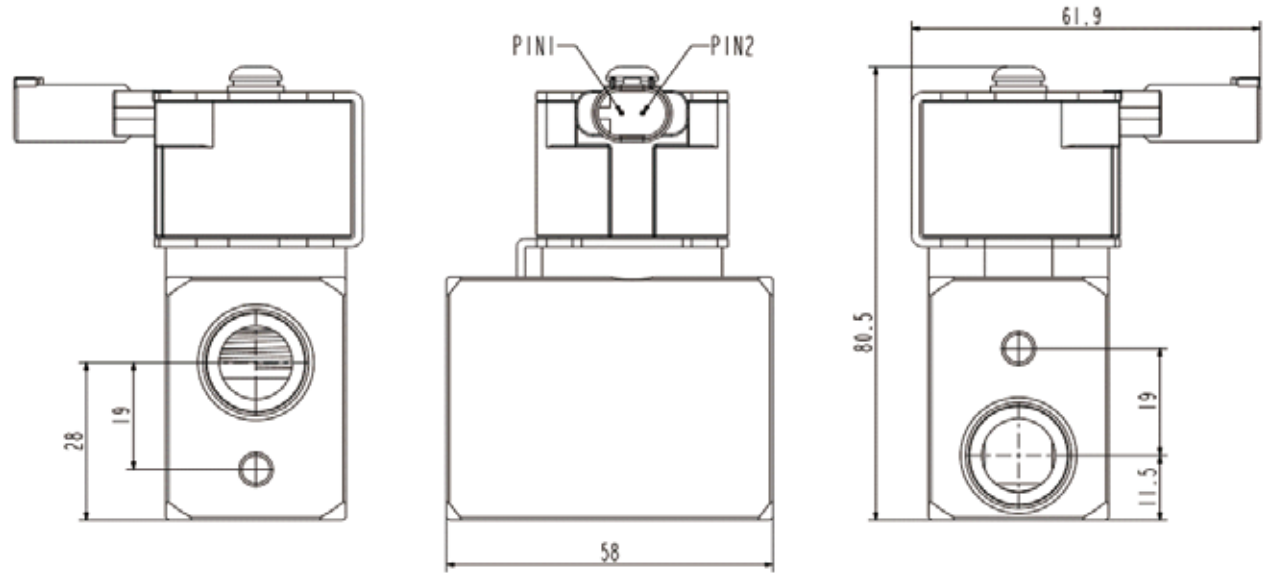
系列 Series	口径 Bore	名义容量(kW) Theoretical Capacity		最大工作压力 Maximum Work Pressure	最大工作压差 Maximum Working Pressure Difference	动作型式 Operating Mode
		R134a/ R1234yf	R290	MPa	MPa	
CBFH2-06*	6.0	10.0	12.0	3.5	2.5	常闭式电磁阀 Normally closed solenoid valve
CBFH2-08*	8.0	14.8	17.8			
CBFH2-10*	10.0	27.9	33.6			
CBFH2-12*	12.0	34.8	41.9			
CKFH2-06*	6.0	10.0	12.0			常开式电磁阀 Normally open solenoid valve
CKFH2-08*	8.0	14.8	17.8			
CKFH2-10*	10.0	27.9	33.6			
CKFH2-12*	12.0	34.8	41.9			

注：系统工况：冷凝温度25℃；蒸发温度-10℃。
Note: System operating conditions: condensation temperature 25 °C; Evaporation temperature -10 °C.

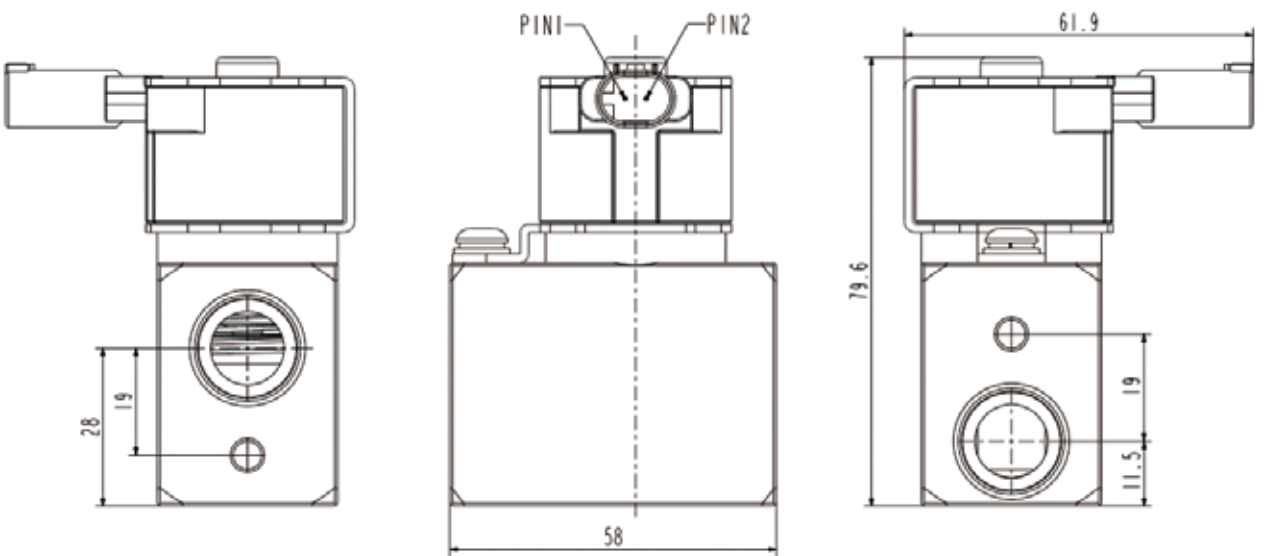
外形尺寸

External Dimension

The series of CBFH2 CBFH2系列



The series of CKFH2 CKFH2系列





H3系列电磁阀

H3 SERIES SOLENOID VALVES

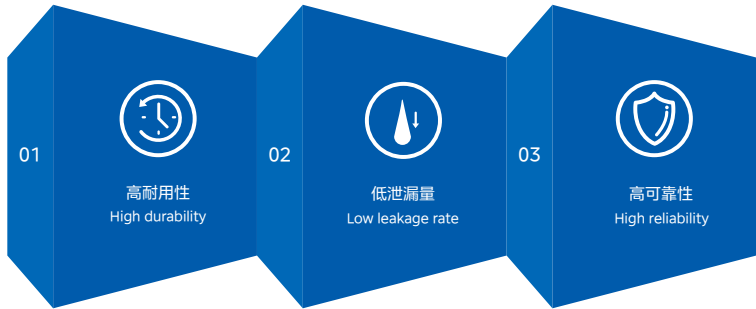


产品应用与特点

Product application and features

H3系列电磁阀主要应用于商用空调制冷系统中，实现制冷剂流路的自动通断。

The H3 series solenoid valve is mainly used in the refrigeration system of commercial air conditioning to achieve automatic on-off of refrigerant flow path.



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R134a/R1234yf/R290/R410A/R454B
◆ 口径范围 Caliber Range	6mm~12mm
◆ 冷媒温度 Refrigerant Temperature	-40℃~+125℃
◆ 环境温度 Ambient Temperature	-40℃~+125℃
◆ 环境湿度 Ambient Humidity	≤98%RH
◆ 工作压力 Working Pressure	3.5MPa
◆ 工作压差 Operating differential pressure	≥1.13MPa (AC21.6V, 20℃)
◆ 内泄漏量 Internal leakage flowrate	≤30ml/min
◆ 安装方式 Installation Method	阀体通过自身螺纹固定，线圈与阀体装配后通过螺钉组件固定。 The valve body is fixed by its own threads, and the coil and valve body are fixed by screw components after assembly.

电气参数 ELECTRICAL PARAMETER		
◆ 额定电压 Rated Voltage		AC24V
◆ 电压范围 Voltage Range		AC21.6V~AC26.4V
◆ 额定电阻 Rated Resistance		13.8Ω±5%Ω(20℃)
◆ 绝缘电阻 Insulation Resistance		≥100MΩ
◆ 绝缘等级 Insulation Class		A级 Grade A
◆ 防护等级 Protection Class		IP67

技术参数

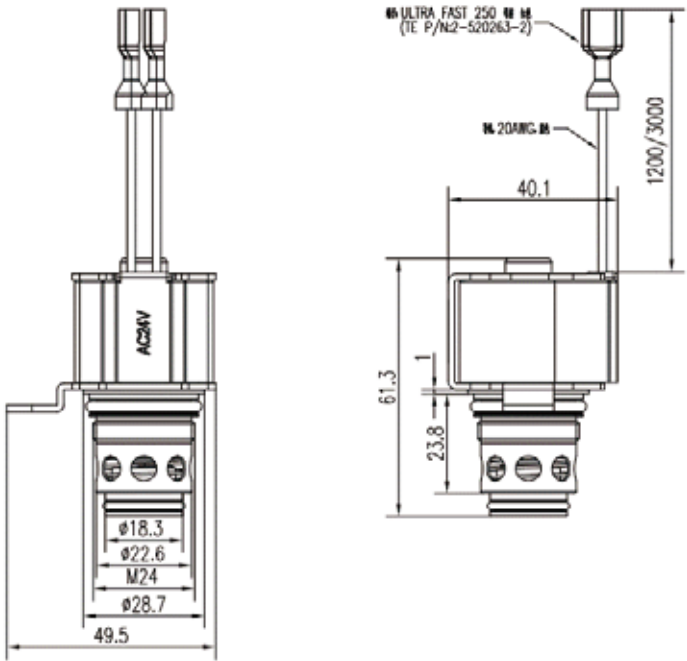
Technical Parameter

系列 Series	口径 Bore	名义容量(kW) Theoretical Capacity			最大工作压力 Maximum Work Pressure	最大工作压差 Maximum Working Pressure Difference	动作型式 Operating Mode
-	mm	R134a/R1234yf	R290	R410A/R454B	MPa	MPa	
CKFH3-06*	6.0	10.0	12.0	10.7	3.5	2.5	常开式电磁阀 Normally open solenoid valve
CKFH3-08*	8.0	14.8	17.8	16.1			
CKFH3-10*	10.0	27.9	33.6	29.6			
CKFH3-12*	12.0	34.8	41.9	36.9			

注：系统工况：冷凝温度25℃；蒸发温度-10℃。
Note: System operating conditions: condensation temperature 25 °C; Evaporation temperature -10 °C.

外形尺寸

External Dimension





技术参数
Technical Parameter

系列 Series	口径 Caliber	名义容量(kW) Nominal Capacity		最大工作压力 Maximum Work Pressure	最低开阀压差 Minimum Valve Opening Pressure Difference	开阀流量 Open Valve Flow Rate
-	mm	R134a/R1234yf	R290	MPa	KPa	ml/min@10KPa
CDFQ1-10*	10.0	253.4	329.5	3.1	≤10	≥500

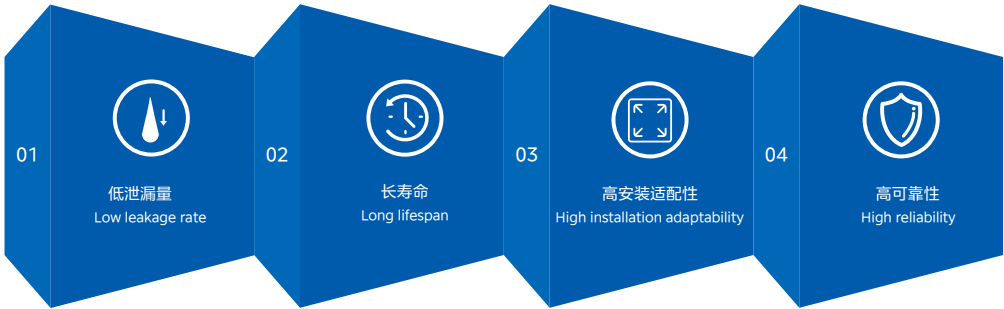
注：额定工况：冷凝温度38℃；蒸发温度5℃；过冷度0℃；过热度0℃。
Note: Rated operating conditions: condensing temperature 38 °C; Evaporation temperature 5 °C; Subcooling degree 0 °C; Overheating degree 0 °C.



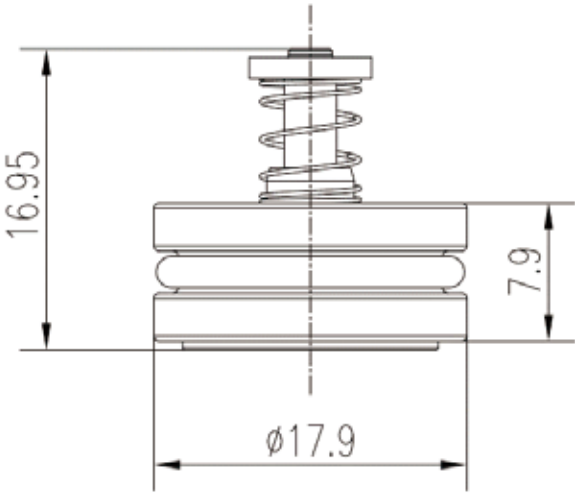
产品应用与特点
Product application and features

Q1系列单向阀主要应用于车用热管理系统中，采用一体嵌入管道式安装，且可以基于系统实际需求多方向安装，进而实现管路系统中正向流通，反向截止，满足系统制冷或制热不同运行模式的需求。

The Q1 series one-way valve is mainly used in automotive thermal management systems, adopting an integrated embedded pipeline installation, and can be installed in multiple directions based on the actual needs of the system, thereby achieving forward flow and reverse shut-off in the pipeline system, meeting the requirements of different operating modes of system cooling or heating.



外形尺寸
Overall Dimension



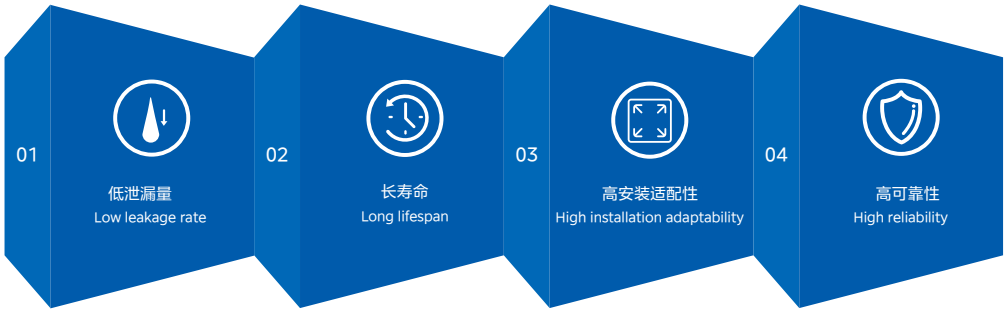
基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Suitable Refrigerant	R134a/R1234yf/R290
◆ 口径范围 Caliber Range	10mm
◆ 冷媒温度 Refrigerant Temperature	-40℃~+125℃
◆ 环境温度 Ambient Temperature	-40℃~+125℃
◆ 工作压力 Work Pressure	3.1MPa
◆ 开阀压差 Valve opening pressure difference	≤10KPa
◆ 安装方式 Installation Method	一体嵌入管道内安装，采用无耳卡簧限位固定 One piece embedded installation in the pipeline, using ear free snap spring limit fixation



产品应用与特点
Product application and features

S1系列单向阀主要应用于车用热管理系统中，采用插接式安装方式，实现管路系统中冷媒正向流通、反向截止功能，满足系统制冷或制热不同运行模式的需求。

The S1 series one-way valve is mainly used in automotive thermal management systems, adopting a plug-in installation method to achieve forward refrigerant flow and reverse shut-off functions in pipeline systems, meeting the needs of different operating modes of system cooling or heating.



基本规格 BASIC SPECIFICATION	
◆ 适用冷媒 Applicable Refrigerant	R134a/R1234yf/R290
◆ 口径范围 Caliber Range	12~16mm
◆ 冷媒温度 Refrigerant Temperature	-40℃~+125℃
◆ 环境温度 Ambient Temperature	-40℃~+125℃
◆ 工作压力 Work Pressure	3.1MPa
◆ 开阀压差 Valve opening pressure difference	≤10KPa
◆ 安装方式 Installation Method	插接式安装方式，采用卡簧限位固定 Plug in installation method, using snap spring limit fixation

技术参数
Technical Parameter

系列 Series	口径 Caliber	名义容量(kW) Nominal Capacity		最大工作压力 Maximum Work Pressure	最低开阀压差 Minimum Valve Opening Pressure Difference	开阀流量 Open Valve Flow Rate
		R134a/ R1234yf	R290			
CDFS1-12	12	217.2	282.4	3.1	≤10	≥500
CDFS1-14*	14	253.4	329.5			
CDFS1-16*	16	648.8	843.3			

注：额定工况：冷凝温度38℃；蒸发温度5℃；过冷度0℃；过热度0℃。
Note: Rated operating conditions: condensing temperature 38 °C; Evaporation temperature 5 °C; Subcooling degree 0 °C; Overheating degree 0 °C.

外形尺寸
External Dimension

